

ADVANCED PREDICTIVE ANALYTICS

STUDENT LABORATORY INSTRUCTION MANUAL

Experiment 06

Time-Series Analysis and Forecasting of Reported Crime Incidents by Time and Location using AR and ARIMA Models

From incident-level public-safety records to leakage-safe temporal forecasting

Manual element	Specification
Scheduled laboratory duration	3 hours
Primary methods	Autoregressive (AR) and ARIMA models
Primary forecasting unit	Aggregated crime-incident counts by day/week/month for a selected location
Core evaluation	Chronological holdout; MAE/RMSE; residual diagnostics
Advanced extension	SARIMA/SARIMAX, rolling-origin backtesting, multiple-location comparison, count-aware/neural forecasting
Recommended environment	Python 3.10+ in Jupyter Notebook / Google Colab
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Semester	Fall Semester 2026-2027

Important interpretation boundary

The models forecast counts of reported incidents, not actual underlying crime prevalence and not individual criminal behavior. Forecasts must not be used for person-level profiling or autonomous policing decisions.

How to Use This Manual

- Complete the pre-lab reading and download the instructor-approved data extract before the scheduled session.
- The 3-hour core focuses on constructing one clean location-level crime-count time series, diagnosing stationarity, fitting AR and ARIMA, and evaluating on future observations.
- Do not randomly shuffle time-series data. Model selection must use only information available before the test period.
- Complete SARIMA/SARIMAX, multiple-city, rolling-origin and neural forecasting only after the core workflow is reproducible.
- Document dataset version/date, filters, location, aggregation frequency, time window and every transformation so that another student can reproduce the series.

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1. Laboratory Title and Positioning

Laboratory title: Time-Series Analysis and Forecasting of Reported Crime Incidents by Time and Location using AR and ARIMA Models

This experiment treats public crime-report data as a temporal count-forecasting problem. Incident-level records are first aggregated into a regular time series (for example weekly counts) for one selected geographic unit such as a police district, borough or neighborhood. Students then build an autoregressive baseline and an ARIMA model, compare future-period forecasts, and examine whether model assumptions and residuals are defensible.

Location is not inserted into a univariate ARIMA model as an arbitrary numeric feature. In the core experiment, location defines which time series is modeled. Advanced learners may compare several location-specific models or use SARIMAX with justified exogenous covariates.

2. Aim and Objectives

2.1 Aim

To analyze reported crime incidents over time and across selected locations, and to build leakage-safe AR and ARIMA forecasting models for future incident counts.

2.2 Objectives

- Convert incident-level records into a regular daily, weekly or monthly crime-count series.
- Examine temporal patterns, trend, seasonality, autocorrelation and location differences.
- Apply chronological training/test separation and avoid random shuffling.
- Assess stationarity using plots and the Augmented Dickey-Fuller (ADF) test.
- Use ACF/PACF diagnostics to motivate AR/ARIMA orders.
- Fit and compare a naive seasonal/last-value benchmark, an AR model and an ARIMA model.
- Evaluate forecasts using MAE and RMSE and inspect residual autocorrelation.
- Interpret forecasts conservatively in the context of reported incidents and data-quality limitations.
- Record all filtering, aggregation, modeling and evaluation choices reproducibly.

3. Learning Outcomes

- Explain why time-series forecasting differs from ordinary shuffled supervised learning.
- Distinguish stationarity, trend, seasonality and autocorrelation.
- Construct a valid regular time series from timestamped incident records.
- Fit autoregressive and ARIMA models using statsmodels.
- Interpret AR order p , differencing order d and moving-average order q .
- Use ACF/PACF, ADF and residual diagnostics appropriately.
- Perform chronological holdout evaluation and compare against a simple baseline.
- Analyze how forecasts differ by district/borough/neighborhood without making person-level inferences.
- Discuss reporting bias, policy changes, missingness and distribution shift in crime data.

4. Problem Statement

A public-safety agency maintains records of reported incidents containing occurrence date/time and geographic information. The objective is to aggregate the incidents into a time series for a selected location and forecast future incident counts using AR and ARIMA techniques.

1. Select an official public crime dataset and document provenance.
2. Choose one defensible spatial unit (e.g., district, borough or neighborhood).
3. Choose a regular forecasting frequency (weekly recommended for the 3-hour core).
4. Construct the count series without using future information.
5. Fit a simple benchmark, AR and ARIMA.
6. Forecast a held-out future period and compare errors.
7. Analyze residuals and model limitations.
8. Optionally compare the same procedure across multiple locations.

5. Public-Safety Context and Technical Framing

Concept	Correct interpretation
Observed target	Number of reported/recorded incidents in a defined place and time interval.
Forecasting unit	One regular time interval for one defined geographic unit.
Location	Defines a separate time series in the core; not a person-level risk feature.
Forecast	Conditional estimate based on historical reporting patterns; not proof of future crime.
Deployment boundary	Decision-support/academic forecasting only; no individual prediction or autonomous resource allocation.

6. Scope and Three-Hour Schedule

Time	Core activity	Expected evidence
0-20 min	Load verified extract; audit date/location semantics; document filter, period, frequency and governance assumptions	Dataset manifest + selected location + time definition
20-45 min	Aggregate to weekly counts; create a complete regular index; inspect missing periods, trend, seasonality and outliers	Regular time series + EDA plots
45-65 min	Create chronological train/validation/test windows; implement seasonal-naive or naive baseline	Frozen temporal split + baseline metrics
65-90 min	ADF test; inspect training-only ACF/PACF; fit and interpret AutoReg AR(p)	Stationarity evidence + AR coefficients/forecast
90-120 min	Fit a small ARIMA candidate set chosen from training diagnostics; compare AIC/BIC on training/validation evidence	Candidate table + selected ARIMA specification
120-145 min	Perform a short rolling-origin/walk-forward	Fold-wise MAE/RMSE + mean/SD

Time	Core activity	Expected evidence
	validation on training/validation data; compare AR versus ARIMA stability	
145-165 min	Evaluate the selected models once on the locked future test period; plot forecasts and prediction intervals where available	Locked-test MAE/RMSE + forecast plot
165-175 min	Inspect residual ACF and Ljung-Box results; identify systematic failures and count-data limitations	Residual diagnostic summary
175-180 min	Replicate the selected model on one second location or record the extension setup; save artifacts and obtain check-off	Second-location evidence or extension plan + saved outputs

7. Verified Public Datasets

The following datasets were verified as publicly accessible official/open-data sources in August 2026. Students should use an instructor-frozen extract for classroom reproducibility and record the access date and filters.

Dataset	Verified source and scope	Why suitable	Recommended role
D1 - Chicago Crimes - 2001 to Present	City of Chicago Data Portal / Chicago Police Department. Reported incidents from 2001 to present (minus most recent 7 days), with date and geographic fields.	Long history, millions of records, police district/community area fields; excellent for weekly/monthly location series.	Preferred core dataset
D2 - NYPD Complaint Data Historic / Current	NYC Open Data / NYPD. Historic valid felony, misdemeanor and violation complaints plus current-year data; occurrence date and borough/precinct/geospatial fields.	Large multi-borough dataset for replication and cross-location analysis.	Benchmark/extension
D3 - SFPD Police Department Incident Reports: 2018 to Present	DataSF / San Francisco Police Department. Incident reports since Jan 1, 2018, updated daily, with datetime and geographic categories.	Modern daily-updated city dataset; useful for domain-shift and neighborhood/district comparison.	Replication/extension

Chicago Crimes - 2001 to Present: <https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2> — Official Chicago Police Department open-data source.

NYPD Complaint Data Historic: <https://data.cityofnewyork.us/Public-Safety/NYPD-Complaint-Data-Historic/qgea-i56i> — Official NYC Open Data historical complaint dataset.

NYPD Complaint Data Current (Year To Date): <https://data.cityofnewyork.us/Public-Safety/NYPD-Complaint-Data-Current-Year-To-Date-/5uac-w243> — Official current-period companion dataset.

SFPD Incident Reports: 2018 to Present: <https://data.sfgov.org/Public-Safety/Police-Department-Incident-Reports-2018-to-Present/wg3w-h783> — Official DataSF incident-report dataset.

7.1 Dataset-selection rule

- For the scheduled core, use one frozen extract from D1 unless the instructor specifies otherwise.

- Do not download millions of rows during the 3-hour slot. The instructor should provide a filtered CSV or parquet file containing the selected years and required columns.
- For fair comparisons, keep the same time window, frequency and target definition when comparing locations or models.
- Do not claim that all cities use identical crime definitions or reporting processes; cross-city comparisons require a documented harmonization step.

8. Data Schema and Governance

Field type	Examples	Use
Time	incident date/time, report date	Construct the time index; prefer occurrence date when appropriate and documented.
Location	district, precinct, borough, community area, neighborhood	Filter or group the series; avoid exact sensitive addresses.
Incident type	primary type, offense description, incident category	Optional category-specific time series.
Identifiers	case number, row ID	Deduplication/audit only; never predictive features.
Coordinates	latitude/longitude	Optional mapping; aggregate before publication and respect source privacy controls.

8.1 Governance checklist

- Document source agency, dataset URL, access date, licence/terms, update frequency and known disclaimers.
- Record whether the chosen date is occurrence date, report date or another timestamp.
- Remove exact victim/person identifiers and avoid publishing precise sensitive locations.
- Document duplicate-record handling and whether one incident can generate multiple rows.
- Distinguish reported incidents from actual prevalence; reporting propensity can vary by location and time.
- Note policing practices, legal definitions, policy changes and data-system changes that can create structural breaks.

9. Theoretical Background

9.1 Time series and autocorrelation

A time series is an ordered sequence y_t observed at regular time intervals. Unlike ordinary tabular data, temporal order is part of the information. Autocorrelation measures dependence between y_t and its lagged values.

9.2 Autoregressive model AR(p)

An AR(p) model predicts the current value from p previous observations:

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t$$

The lag order p can be motivated using PACF and validated using training-only information criteria or forecasting performance.

9.3 ARIMA(p,d,q)

ARIMA combines autoregression, differencing and a moving-average error component. The differencing order d is used to obtain a more stationary series when appropriate.

ARIMA(p, d, q): AR order p + differencing order d + MA order q

9.4 Stationarity

A weakly stationary series has approximately stable mean, variance and autocovariance over time. Visual inspection and the Augmented Dickey-Fuller (ADF) test are useful diagnostics, but neither should be used mechanically. Differencing can remove trend, while seasonal structure may require seasonal modeling.

9.5 ACF and PACF

- ACF: correlation between a series and its lagged values; useful for detecting persistence and suggesting MA/seasonal structure.
- PACF: correlation at a lag after controlling for intermediate lags; often useful for suggesting AR order.

9.6 Forecast errors

MAE = mean(|y - $\hat{y}$ |) — interpretable in incidents per interval.

RMSE = sqrt(mean((y - $\hat{y}$)²)) — penalizes large errors more strongly.

MAPE is not recommended as the primary metric when crime counts can be zero or near zero. Report MAE/RMSE and optionally sMAPE with caveats.

9.7 Residual diagnostics

A defensible model should leave residuals with little predictable temporal structure. Inspect residual plots, residual ACF and the Ljung-Box test. A low Ljung-Box p-value at important lags suggests remaining autocorrelation and model inadequacy.

10. End-to-End Workflow

9. Define forecasting target, location, frequency and horizon.
10. Load the official/instructor-frozen incident extract.
11. Parse timestamps and audit duplicates.
12. Filter one location and optional crime category.
13. Aggregate incident records to regular counts.
14. Reindex missing time intervals to zero only when zero truly means no recorded incidents, not missing data.
15. Plot trend/seasonality and inspect outliers/structural breaks.
16. Create chronological train/test split.
17. Fit a naive baseline.
18. Run ADF and inspect ACF/PACF on training data only.
19. Fit AutoReg and ARIMA candidates.
20. Forecast the fixed test horizon.
21. Compare MAE/RMSE and residual diagnostics.
22. Interpret by time/location and document limitations.
23. Save predictions, model orders, diagnostics and environment metadata.

11. Software and Libraries

Package	Purpose
pandas / numpy	Data loading, aggregation, time index and numerical operations
matplotlib	Time-series and diagnostic visualization
statsmodels	AutoReg, ARIMA, ADF, ACF/PACF, Ljung-Box
scikit-learn	MAE/RMSE utilities
joblib / json	Optional artifact persistence and configuration metadata

```
# Core environment
!pip -q install pandas numpy matplotlib statsmodels scikit-learn joblib
```

12. Step-by-Step Core Procedure

12.1 Configure the experiment

```
from pathlib import Path
import json, platform, sys, time
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.metrics import mean_absolute_error, mean_squared_error
from statsmodels.tsa.ar_model import AutoReg
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.stattools import adfuller, acf, pacf
from statsmodels.stats.diagnostic import acorr_ljungbox

SEED = 42
np.random.seed(SEED)
OUT = Path('lab06_outputs')
OUT.mkdir(exist_ok=True)

CONFIG = {
    'dataset': 'Chicago Crimes - instructor frozen extract',
    'date_col': 'Date',
    'location_col': 'District',
    'location_value': 1,          # change to instructor-assigned district
    'frequency': 'W-MON',        # weekly series
    'test_periods': 12,
    'seed': SEED
}
print(CONFIG)
```

12.2 Load and validate data

```
DATA_PATH = 'crime_extract.csv' # instructor-provided path
df = pd.read_csv(DATA_PATH)

required = {CONFIG['date_col'], CONFIG['location_col']}
```



```

missing = required - set(df.columns)
assert not missing, f'Missing required columns: {missing}'

df[CONFIG['date_col']] = pd.to_datetime(df[CONFIG['date_col']], errors='coerce')
df = df.dropna(subset=[CONFIG['date_col']]).copy()
print(df.shape)
print(df[[CONFIG['date_col'], CONFIG['location_col']]].head())

```

12.3 Construct a regular location-level series

```

loc = df[df[CONFIG['location_col']].astype(str) == str(CONFIG['location_value'])].copy()
assert len(loc) > 0, 'Selected location has no observations.'

# Count incident rows by regular time period.
y = (loc.set_index(CONFIG['date_col'])
      .resample(CONFIG['frequency'])
      .size()
      .rename('incidents')
      .asfreq(CONFIG['frequency'], fill_value=0))

assert y.index.is_monotonic_increasing
assert y.index.is_unique
assert y.isna().sum() == 0
print(y.describe())

y.plot(figsize=(10,4), title='Reported incident counts - selected location')
plt.ylabel('Incidents per period')
plt.tight_layout(); plt.show()

```

12.4 Chronological split and naive baseline

```

H = CONFIG['test_periods']
assert len(y) > 3*H, 'Series is too short for the chosen holdout.'
train, test = y.iloc[:-H], y.iloc[-H:]

# Persistence / recent-mean baseline. Choose using training data only.
naive_pred = np.repeat(train.iloc[-1], len(test))

def score(y_true, y_pred):
    return {
        'MAE': mean_absolute_error(y_true, y_pred),
        'RMSE': mean_squared_error(y_true, y_pred)**0.5
    }
print('Naive:', score(test, naive_pred))

```

12.5 Stationarity and lag diagnostics

```

adf_stat, adf_p, *_ = adfuller(train)
print(f'ADF statistic={adf_stat:.3f}, p-value={adf_p:.4f}')

fig, ax = plt.subplots(1, 2, figsize=(10,3.5))
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
plot_acf(train, lags=min(30, len(train)//4), ax=ax[0])
plot_pacf(train, lags=min(30, len(train)//4), ax=ax[1], method='ywm')
plt.tight_layout(); plt.show()

```

12.6 Fit AR model

```
AR_LAGS = 4 # justify from training PACF/CV; instructor may fix for core
ar_model = AutoReg(train, lags=AR_LAGS, old_names=False, trend='ct').fit()
ar_pred = ar_model.predict(start=len(train), end=len(train)+len(test)-1, dynamic=False)
print('AR:', score(test, ar_pred))
```

12.7 Fit ARIMA model

```
# Keep candidate search intentionally small in the 3-hour core.
candidates = [(1,0,0), (2,0,0), (1,1,1), (2,1,1)]
rows=[]
for order in candidates:
    try:
        m = ARIMA(train, order=order).fit()
        rows.append({'order':order, 'AIC':m.aic, 'model':m})
    except Exception as e:
        print('Failed', order, e)

ranked = sorted(rows, key=lambda z: z['AIC'])
best = ranked[0]
print('Training-selected order:', best['order'], 'AIC=', best['AIC'])
arima_model = best['model']
arima_pred = arima_model.forecast(steps=len(test))
print('ARIMA:', score(test, arima_pred))
```

12.8 Compare forecasts and residuals

```
results = pd.DataFrame([
    {'Model':'Naive', **score(test, naive_pred)},
    {'Model':f'AR({AR_LAGS})', **score(test, ar_pred)},
    {'Model':f'ARIMA{best["order"]}', **score(test, arima_pred)}
]).sort_values('MAE')
print(results)
results.to_csv(OUT/'model_comparison.csv', index=False)

pred_df = pd.DataFrame({
    'actual': test,
    'naive': naive_pred,
    'AR': np.asarray(ar_pred),
    'ARIMA': np.asarray(arima_pred)
}, index=test.index)
pred_df.plot(figsize=(10,4), marker='o', title='Future holdout: actual vs forecast')
plt.ylabel('Reported incidents'); plt.tight_layout(); plt.show()

resid = arima_model.resid.dropna()
print(acorr_ljungbox(resid, lags=[min(10, max(2,len(resid)//10))], return_df=True))
```

12.9 Save manifest and interpretation

```
manifest = {
    **CONFIG,
    'n_total_periods': int(len(y)),
    'n_train': int(len(train)),
    'n_test': int(len(test)),
    'selected_arima_order': list(best['order']),
    'python': sys.version,
    'platform': platform.platform()
}
(OUT/'manifest.json').write_text(json.dumps(manifest, indent=2, default=str))
```

```
pred_df.to_csv(OUT/'test_predictions.csv')
```

13. AR and ARIMA Models

Model	Purpose	Core requirement	Interpretation caution
Naive baseline	Reference forecast using recent history	Mandatory	A complex model must beat this baseline to justify itself.
AR(p)	Models persistence through lagged observed counts	Mandatory	Trend/seasonality can make coefficients misleading if untreated.
ARIMA(p,d,q)	Adds differencing and moving-average error structure	Mandatory	Order must be selected from training evidence, not test results.
SARIMA/SARIMAX	Adds seasonal terms/exogenous variables	Advanced	Use only with enough seasonal cycles and justified covariates.

14. Time-Series Evaluation and Diagnostics

14.1 Required metrics

- MAE for incident-count error in original units.
- RMSE to emphasize larger misses.
- AIC/BIC only as training-model selection aids, not as substitutes for future forecast evaluation.
- Residual ACF and Ljung-Box for remaining autocorrelation.

14.2 Evaluation protocol

- Use the oldest observations for training and newest observations for testing.
- Never randomize or shuffle the temporal order.
- All transformations, lag choices and model orders must be chosen without consulting the final test values.
- For advanced learners, use rolling-origin evaluation rather than one lucky split.

14.3 Count-data caveat

AR/ARIMA are continuous Gaussian-style forecasting models. Crime series are non-negative counts and may be overdispersed or zero-heavy. Negative forecasts should be reported as a model limitation; do not silently clip them without documenting the transformation. Advanced learners may compare log1p-transformed ARIMA or count-aware models.

15. Time-and-Location Analysis

The syllabus includes both time and location. The recommended design is to build comparable time series for several geographic units rather than encode location as a number inside a univariate ARIMA.

Analysis	Core/Advanced	Method
One-location forecast	Core	Select one district/borough; weekly series; AR vs ARIMA.

Analysis	Core/Advanced	Method
Two-location comparison	Optional core if time permits	Same frequency/window; compare trend, volatility and errors separately.
Many-location replication	Advanced	Fit same protocol per location; summarize MAE/RMSE and data support.
Map + forecast summary	Advanced	Aggregate location forecasts; display counts/errors without person-level risk scoring.
Exogenous/location context	Advanced	SARIMAX with justified weather/holiday/event variables or hierarchical methods; do not treat location code as ordinal.

16. Advanced Learner Extension

16.1 SARIMA for seasonality

Weekly crime counts may show annual or shorter seasonal cycles. Advanced learners may compare ARIMA with SARIMA using a justified seasonal period and sufficient historical cycles.

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
# Example only: choose seasonal period from domain/diagnostics, not blindly.
sarima = SARIMAX(train, order=(1,1,1), seasonal_order=(1,0,1,52),
                  enforce_stationarity=False, enforce_invertibility=False).fit(dispatch=False)
sarima_pred = sarima.forecast(len(test))
print(score(test, sarima_pred))
```

16.2 Rolling-origin backtesting

Use multiple forecast origins to measure variability across time. Report mean/SD of MAE/RMSE and avoid claims based on a single holdout.

16.3 SARIMAX with exogenous features

- Calendar: holidays, month, school term, major scheduled events.
- Weather: temperature/precipitation only when matched causally and temporally.
- Policy/system indicators: known data-system changes or reporting-policy changes.

Every exogenous feature must be known at forecast time or itself forecastable. Do not leak future realized information.

16.4 Advanced neural forecasting (optional research comparison)

For advanced learners only, compare an LSTM/GRU or Temporal Fusion Transformer against the statistical baseline using the identical chronological split or rolling-origin folds. The advanced model is useful only if the gain is reproducible and justifies additional compute/complexity. AR/ARIMA remain the assessed conceptual focus.

16.5 Count-aware alternatives (research discussion)

Students may investigate Poisson/negative-binomial autoregressive models or state-space count models when the Gaussian assumptions of ARIMA are problematic. This is an extension, not a replacement for understanding AR/ARIMA.

17. Visualization Requirements

- Raw incident-count series with train/test boundary.
- Rolling mean/variance or seasonal aggregation for exploratory analysis.
- ACF and PACF on training data.
- Actual versus naive/AR/ARIMA forecasts on the locked future period.
- ARIMA residual time plot and residual ACF.
- Core/extension: compare a second location using identical time scale, aggregation frequency and forecast horizon.
- Advanced: rolling-origin error distribution and seasonal decomposition.

Every figure must include a title, axis labels, units/frequency, legend where needed, and 2-3 sentences interpreting the pattern without implying causality.

18. Required Experiments

18.1 Core (3 hours)

- One verified dataset and one selected location.
- Weekly (or instructor-approved) count series.
- Chronological holdout.
- Naive baseline.
- ADF + ACF/PACF diagnostics.
- AR model.
- At least three small ARIMA candidate orders selected using training information.
- Locked-test MAE/RMSE.
- Residual diagnostics.
- Five-sentence interpretation of time/location limitations.
- Short rolling-origin/walk-forward validation using training/validation history only; report fold-wise MAE/RMSE and variability.
- Prediction-interval interpretation where the fitted model provides intervals; discuss empirical coverage without treating intervals as guarantees.
- Replication on one second location using the same frequency, split logic and evaluation protocol, or completion as the first extension task if data preparation is slow.

18.2 Advanced extension

- Replicate on D2 or D3.
- Compare at least three locations with same protocol.
- SARIMA seasonal model.
- Expanded rolling-origin backtesting across multiple forecast origins, horizons and locations, with uncertainty comparison.
- Category-specific series (e.g., theft/assault) with label harmonization documented.
- Structural-break/temporal-drift analysis.
- Optional SARIMAX with valid exogenous features.
- Optional LSTM/GRU/TFT comparison with identical evaluation.

19. Result-Table Templates

Dataset	Agency/source	Time span used	Location field	Aggregation	Rows before filter	Series periods

Model	Order/lags	Train AIC/BIC	Test MAE	Test RMSE	Ljung-Box p	Notes
Naive	-	-			-	
AR						
ARIMA						

Location	Periods	Mean count	Std. dev.	Best model	MAE	RMSE	Interpretation

Advanced experiment	Baseline	Advanced model	Metric change	Runtime change	Conclusion

20. Discussion Questions

1. Why is a random train/test split invalid for this experiment?
2. What is the difference between an incident occurrence date and report date, and why does it matter?
3. Why does location define separate time series in the core rather than an ordinal numeric ARIMA feature?
4. What does stationarity mean in practical terms for weekly crime counts?
5. What do p, d and q represent in ARIMA?
6. Why should AIC not be used as the only evidence that one model forecasts better?
7. What does residual autocorrelation tell you?
8. Why can MAPE be unstable for count series?
9. How can holidays, weather, policy changes or reporting-system changes cause distribution shift?
10. Why might a crime-count series require seasonal modeling?
11. What are the consequences of fitting an ARIMA model that produces negative counts?
12. Why is forecasting reported incidents not the same as forecasting actual crime prevalence?
13. What risks arise if forecast maps are interpreted as person-level risk maps?
14. How would you compare two locations fairly?

15. When would an advanced neural model be justified over ARIMA?

21. Viva Questions

Viva question	Expected key point
What is an AR model?	A regression of the current value on its own previous lagged values.
What does AR(p) mean?	p is the number of autoregressive lags.
What does ARIMA stand for?	Autoregressive Integrated Moving Average.
What does d represent?	The number/order of non-seasonal differences.
What does q represent?	The order of the moving-average error component.
Why chronological split?	Future observations must not influence past model fitting or tuning.
What is stationarity?	Approximately stable distributional properties such as mean/variance/autocovariance over time.
What is the ADF test?	A unit-root test used as one diagnostic for stationarity.
What is ACF?	Correlation of a series with its own lagged values.
What is PACF?	Lag correlation after controlling for intermediate lags.
What is differencing?	Subtracting previous values to reduce nonstationary trend.
Why a naive baseline?	To verify that model complexity produces meaningful forecasting benefit.
What is MAE?	Mean absolute forecast error in original units.
What is RMSE?	Root mean squared error; more sensitive to large misses.
Why not rely on MAPE?	Zeros/near-zeros make percentage errors undefined or unstable.
What is AIC?	An in-sample likelihood-complexity criterion useful for candidate comparison.
What is residual white noise?	Residuals with little systematic temporal dependence.
What is Ljung-Box?	A test for residual autocorrelation over selected lags.
What is SARIMA?	ARIMA extended with seasonal AR, differencing and MA terms.
What is SARIMAX?	SARIMA with exogenous regressors.
What is rolling-origin evaluation?	Repeated future forecasts from successive historical cutoffs.
What is temporal leakage?	Using information from after the forecast origin during training/tuning.

Viva question	Expected key point
Why aggregate incidents?	ARIMA needs a regular numeric series rather than irregular incident rows.
What is a structural break?	A persistent change in the data-generating/reporting process.
Why is crime data ethically sensitive?	Reported incidents reflect human behavior, reporting, enforcement and policy; misuse can reinforce inequity.

22. Common Mistakes

Mistake	Why incorrect	Correct practice
Random train/test split	Leaks future temporal structure into training.	Use chronological holdout or rolling origin.
Using raw incident rows directly in ARIMA	ARIMA requires a regular numeric series.	Aggregate counts at a fixed frequency.
Treating district ID as a continuous ARIMA regressor	Location codes are labels, not ordered magnitudes.	Build separate series or use a justified multivariate/hierarchical method.
Choosing p,d,q from test performance	Tunes to the final future holdout.	Use training diagnostics/AIC/CV; evaluate test once.
Ignoring missing periods	Irregular spacing breaks time-series assumptions.	Reindex regularly and distinguish true zero from missing data.
Differencing repeatedly until ADF passes	Can over-difference and destroy signal.	Combine diagnostics, theory and forecast validation.
Reporting only AIC	AIC is not out-of-sample forecasting accuracy.	Report MAE/RMSE on future data.
Calling reported incidents “true crime rate”	Reporting and recording processes affect counts.	Use precise language: reported/recorded incidents.
Clipping negative forecasts silently	Hides model mismatch with count data.	Report limitation or use documented transform/count model.
Cross-city direct comparison without harmonization	Offense definitions/reporting systems differ.	Document label and process harmonization.

23. Responsible Analytics and Limitations

- Reported incidents are an administrative measurement, not a complete census of crime.
- Reporting propensity, enforcement intensity, legal definitions and data-system changes can differ across neighborhoods and time.
- A location forecast must not be interpreted as individual propensity, guilt or dangerousness.
- Avoid presenting precise household/address-level forecasts; use aggregate geographic units.
- Do not recommend autonomous patrol allocation or punitive actions from a classroom forecast.

- Analyze model error across locations before making comparative statements.
- Document structural breaks, missingness, delayed reporting and major policy changes.
- If demographic or socioeconomic external data are introduced in an extension, require explicit fairness, privacy and causal-use review.

23.1 Risk register

Risk	Cause	Potential impact	Mitigation
Under-reporting bias	Not all incidents are reported equally	Forecast reflects reporting process as well as events	Use precise language; triangulate if possible
Spatial stigmatization	Area-level counts interpreted as people-level risk	Reputational and fairness harm	Aggregate, contextualize, no person-level inference
Temporal drift	Policy, season, population or reporting system changes	Forecast degradation	Rolling evaluation and monitoring
Data leakage	Random splits/future information	Artificially optimistic errors	Chronological split and provenance checks

24. Submission Guidelines

- RegistrationNumber_Lab06_Crime_AR_ARIMA.ipynb
- RegistrationNumber_Lab06_Report.pdf
- RegistrationNumber_Lab06_Model_Comparison.csv
- RegistrationNumber_Lab06_Test_Predictions.csv
- RegistrationNumber_Lab06_Manifest.json
- figures/ containing required plots
- README.md containing dataset source, filters, location, frequency, execution order and limitations

24.1 Short report structure

24. Aim and forecasting question
25. Dataset provenance and selected location
26. Aggregation/frequency and chronological split
27. Stationarity and lag diagnostics
28. AR and ARIMA models
29. Forecast results
30. Residual/error analysis
31. Time/location interpretation
32. Limitations and responsible-use statement
33. Conclusion

25. Assessment Rubric - 100 Marks

Criterion	Marks	Excellent evidence
Problem framing & target definition	6	Forecast unit, horizon, location and “reported incident” language are correct.

Criterion	Marks	Excellent evidence
Dataset provenance & governance	8	Official source, access date, filters, time/location fields, duplicates and limitations documented.
Time-series construction	12	Correct timestamp parsing, aggregation, regular index, missing-period treatment and saved series.
Chronological split & leakage control	10	No shuffle; train/test boundary fixed before modeling; test not used for order selection.
EDA, stationarity & ACF/PACF	12	Readable temporal plots, ADF interpretation, ACF/PACF used critically rather than mechanically.
Naive baseline & AR model	10	Baseline and AutoReg implemented correctly; lag choice explained.
ARIMA implementation & order selection	15	Multiple training-selected candidates; p,d,q interpretation; model converges and forecast generated.
Forecast evaluation & residual diagnostics	12	MAE/RMSE, actual-vs-predicted plot, residual ACF/Ljung-Box and evidence-based comparison.
Time/location interpretation	6	Location used appropriately; no unsupported causal/person-level claims.
Responsible analytics & limitations	4	Reporting bias, privacy, spatial stigma and deployment boundary addressed.
Reproducibility & artifacts	3	Notebook runs top-to-bottom; manifest/predictions/figures saved.
Technical communication	2	Concise, accurate report and figures.

Performance bands: 90-100 Exceptional | 80-89 Very Good | 70-79 Good | 60-69 Satisfactory | <60 Insufficient

26. Final Checklist

- ☐ Official/instructor-approved dataset source and access date recorded.
- ☐ Selected location and aggregation frequency documented.
- ☐ Occurrence/report date choice justified.
- ☐ Series is regular, ordered, unique and has no unexplained missing intervals.
- ☐ Chronological train/test split used; no random shuffling.
- ☐ Naive baseline, AR and ARIMA results reported.
- ☐ ARIMA order selected without using final test values.
- ☐ ADF and ACF/PACF diagnostics included.
- ☐ MAE and RMSE reported on the locked future period.
- ☐ Residual diagnostic and Ljung-Box result included.
- ☐ Forecast plot includes actual values and train/test boundary.
- ☐ No person-level or causal crime claim is made.

- ☐ Manifest, predictions, figures and notebook saved.
- ☐ Advanced experiments clearly separated from the core if performed.

27. References and Student-Help Links

Students should record access dates and software versions. Official data portals should be preferred over unofficial mirrors.

Statsmodels - AutoReg: https://www.statsmodels.org/stable/generated/statsmodels.tsa.ar_model.AutoReg.html — Official implementation documentation for autoregressive models.

Statsmodels - ARIMA: <https://www.statsmodels.org/stable/generated/statsmodels.tsa.arima.model.ARIMA.html> — Official ARIMA model documentation.

Statsmodels - Time Series Analysis: <https://www.statsmodels.org/stable/tsa.html> — ADF, ACF/PACF, forecasting and diagnostic tools.

Chicago Crimes - 2001 to Present: <https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2> — Official Chicago Police Department open dataset.

NYPD Complaint Data Historic: <https://data.cityofnewyork.us/Public-Safety/NYPD-Complaint-Data-Historic/qgea-i56i> — Official NYC Open Data historical crime complaints.

SFPD Incident Reports: 2018 to Present: <https://data.sfgov.org/Public-Safety/Police-Department-Incident-Reports-2018-to-Present/wg3w-h783> — Official DataSF police incident reports.

Forecasting: Principles and Practice (online textbook): <https://otexts.com/fpp3/> — Open forecasting reference for time-series decomposition, ARIMA and evaluation.

Appendix A. Executable Notebook Scaffold

The code in Section 12 is the minimum executable scaffold. Students must adapt only the configurable path, column names, location and time window. The instructor should provide the final column mapping for the classroom extract.

A.1 Optional crime-type filter

```
# Example: if the instructor extract contains a category column
CATEGORY_COL = 'Primary Type'
CATEGORY_VALUE = 'THEFT'
if CATEGORY_COL in df.columns:
    loc = loc[loc[CATEGORY_COL].astype(str).str.upper() == CATEGORY_VALUE]
```

A.2 Forecast intervals

```
# ARIMA forecast object with approximate confidence intervals
fc = arima_model.get_forecast(steps=len(test))
mean_fc = fc.predicted_mean
ci = fc.conf_int(alpha=0.05)
print(ci.head())
```

Appendix B. Advanced Backtesting and Model Extensions

B.1 Simple rolling-origin evaluation skeleton

```
def rolling_origins(series, initial, horizon, step):
    origins=[]
    end=initial
    while end + horizon <= len(series):
        origins.append((series.iloc[:end], series.iloc[end:end+horizon]))
        end += step
    return origins

rows=[]
for fold,(tr,te) in enumerate(rolling_origins(y, initial=max(52, len(y)//2), horizon=8, step=8),1):
    try:
        m=ARIMA(tr, order=best['order']).fit()
        pr=m.forecast(len(te))
        rows.append({'fold':fold, **score(te,pr)})
    except Exception as e:
        print('fold failed', fold, e)
print(pd.DataFrame(rows))
```

B.2 Advanced model-complexity question

If an LSTM/GRU/TFT or another advanced model is evaluated, students must report at least three repeated runs or an uncertainty interval, training time, inference time and parameter/model size where feasible. The report must explicitly answer whether the gain over ARIMA is large and stable enough to justify the extra complexity.

Appendix C. Acceptance Tests and Reproducibility

```
# Core acceptance tests
assert y.index.is_monotonic_increasing
assert y.index.is_unique
assert y.isna().sum() == 0
assert len(train) + len(test) == len(y)
assert train.index.max() < test.index.min()
assert len(test) == CONFIG['test_periods']
assert set(['actual', 'naive', 'AR', 'ARIMA']).issubset(pred_df.columns)
assert (OUT/'model_comparison.csv').exists()
assert (OUT/'test_predictions.csv').exists()
assert (OUT/'manifest.json').exists()
print('Core acceptance tests passed.')
```

C.1 Reproducibility record

Item	Record
Dataset name/version/access date	
Source URL	
Instructor extract checksum	
Date column used	

Item	Record
Location column/value	
Crime category filter (if any)	
Aggregation frequency	
Observation window	
Forecast horizon	
AR lags	
ARIMA candidate orders	
Selected ARIMA order	
Python/statsmodels versions	